

SOLDOTNA, AK, USAWMO: **702595**Lat: **60.476N**Lon: **151.034W**Elev: **34**StdP: **100.91**Time Zone: **-9.00 (NAK)**Period: **94-19**WBAN: **26559****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.0	-27.2	-32.9	0.2	-27.2	-31.3	0.2	-24.4	7.2	1.1	5.9	0.1	1.0	100	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	22.4	14.9	20.8	14.0	18.8	12.9	15.6	21.2	14.5	19.2	13.6	17.7	3.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.7	9.2	17.3	12.2	8.9	16.5	11.3	8.4	15.5	43.5	21.4	40.4	19.0	37.9	17.7	20.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	5.7	5.0	-30.9	24.8	4.2	2.3	-34.0	26.4	-36.4	27.8	-38.8	29.1	-41.9	30.8		
(5)			-30.6	16.4	3.6	1.5	-33.2	17.5	-35.3	18.4	-37.4	19.3	-40.0	20.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-10.4	-6.8	-4.6	2.2	7.2	11.1	13.5	12.7	8.7	1.7	-5.7	-8.6	(6)
(7)		DBStd	9.86	8.61	6.69	6.12	3.65	2.66	2.38	1.96	2.23	2.79	4.86	6.42	7.39	(7)
(8)		HDD10.0	3258	633	470	452	234	93	14	1	3	55	257	470	578	(8)
(9)		HDD18.3	6038	891	703	710	484	347	217	152	174	289	514	720	836	(9)
(10)		CDD10.0	264	0	0	0	0	6	47	108	88	16	1	0	0	(10)
(11)		CDD18.3	1	0	0	0	0	0	0	1	1	0	0	0	0	(11)
(12)		CDH23.3	26	0	0	0	0	0	6	13	7	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	0	2	1	0	0	0	0	(13)
(14)	Wind	WSAvg	1.8	1.5	1.5	1.7	2.0	2.4	2.2	1.9	1.7	1.6	1.6	1.5	1.6	(14)
(15)	Precipitation	PrecAvg	594	40	37	26	23	23	32	55	69	94	78	57	57	(15)
(16)		PrecMax	788	95	96	77	67	63	66	115	126	168	206	125	116	(16)
(17)		PrecMin	466	7	1	5	5	4	9	18	13	35	42	1	11	(17)
(18)		PrecStd	80	24	19	18	16	16	15	27	29	37	37	38	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	6.2	8.8	14.8	21.2	24.1	25.0	24.2	18.8	13.7	7.9	5.9	(19)
(20)			MCWB	3.5	2.8	4.1	7.5	11.2	14.9	15.9	16.7	11.9	9.2	5.0	3.6	(20)
(21)		2%	DB	3.8	4.0	7.1	12.2	17.8	21.5	22.4	21.8	17.0	11.2	5.9	3.7	(21)
(22)			MCWB	1.7	1.6	2.9	5.8	9.7	13.4	15.3	15.3	11.2	7.5	3.7	2.1	(22)
(23)		5%	DB	2.4	2.8	5.1	10.4	16.0	19.0	20.6	19.2	15.1	9.6	3.9	2.3	(23)
(24)			MCWB	0.7	0.7	1.6	4.9	8.6	12.3	14.5	13.5	10.3	6.7	2.2	1.0	(24)
(25)		10%	DB	1.1	2.0	3.6	8.7	13.7	17.4	18.6	17.8	13.7	8.0	2.6	1.0	(25)
(26)			MCWB	-0.2	0.3	0.6	4.1	7.8	11.4	13.4	12.7	9.9	5.7	1.3	0.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	3.3	4.9	8.2	12.0	15.8	16.9	17.4	13.0	9.9	5.4	3.9	(27)
(28)			MCDWB	6.0	5.3	7.8	13.9	19.1	23.1	23.2	23.2	16.8	12.8	7.4	5.5	(28)
(29)		2%	WB	1.9	2.2	3.3	6.2	10.2	14.0	15.6	15.7	11.9	8.2	4.0	2.3	(29)
(30)			MCDWB	3.4	3.9	6.3	10.8	16.6	19.9	21.2	20.8	14.9	10.1	5.7	3.5	(30)
(31)		5%	WB	0.8	1.1	2.0	5.2	9.1	12.8	14.7	14.3	11.2	7.3	2.7	0.9	(31)
(32)			MCDWB	2.2	2.6	4.7	9.4	14.7	18.2	19.5	18.5	14.1	9.1	3.9	2.0	(32)
(33)		10%	WB	-0.4	0.1	1.0	4.2	8.2	11.7	13.9	13.3	10.5	6.2	1.4	-0.2	(33)
(34)			MCDWB	0.8	1.6	3.4	8.3	13.2	16.7	17.9	17.0	13.2	8.0	2.5	0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.9	9.7	11.4	10.0	10.6	9.9	8.7	9.1	8.8	8.2	7.5	7.4	(35)
(36)			MCDBR	7.1	8.0	10.0	12.4	15.9	14.9	13.2	12.8	11.3	9.0	6.2	7.0	(36)
(37)			MCWBR	5.9	6.3	7.1	7.6	9.2	8.3	7.3	7.4	7.0	6.0	4.9	5.9	(37)
(38)		5% WB	MCDBR	7.0	7.3	9.0	10.9	13.3	12.9	11.3	10.7	9.1	7.4	6.0	6.7	(38)
(39)			MCWBR	5.8	5.9	6.5	7.0	8.0	7.5	6.6	6.5	6.1	5.2	4.9	5.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.239	0.283	0.301	0.345	0.372	0.362	0.369	0.353	0.326	0.300	0.252	0.226	(40)	
(41)		taud	2.163	2.285	2.322	2.323	2.327	2.437	2.448	2.495	2.551	2.497	2.171	1.980	(41)	
(42)		Ebn at Noon	535	719	831	852	852	869	852	840	808	709	497	348	(42)	
(43)		Edh at Noon	39	64	86	104	112	102	99	88	70	53	38	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.30	0.97	2.50	4.01	5.20	5.43	4.79	3.81	2.47	1.18	0.43	0.15	(44)	
(45)		RadStd	0.04	0.13	0.29	0.41	0.41	0.39	0.46	0.46	0.35	0.12	0.07	0.02	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		+0.68	+2.15	+1.89	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page